vSphere and Virtualization Overview

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Importance

As a vSphere administrator, you must be familiar with the components on which vSphere is based.

You must also understand the following concepts:

- Virtualization, the role of the ESXi hypervisor in virtualization, and virtual machines
- Fundamental vSphere components and the use of vSphere in the software-defined data center
- When to use each one of the user interfaces to administer and manage vSphere environments

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Module Lessons

1. vSphere Virtualization Overview
2. vSphere Virtualization of Resources

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vSphere Virtualization Overview



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Learner Objectives

- Explain basic virtualization concepts
- Describe vSphere
- Describe how vSphere fits in the software-defined data center
- Describe vSphere+
- Recognize the user interfaces for accessing vSphere

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Terminology (1)

Virtualization is associated with several key concepts, products, and features.

Term	Definition	Examples
Operating system	Software designed to allocate physical resources to applications	Microsoft Windows, Linux
Application	Software that runs on an operating system, consuming physical resources	Microsoft Office, Chrome
Hypervisor	Specialized operating system designed to run VMs	ESXi, Workstation, Fusion
Virtual machine	Specialized application that abstracts hardware resources into software	
Guest	The operating system that runs in a VM (also called the guest operating system)	Microsoft Windows, Linux
Host	Physical computer that provides resources to the ESXi hypervisor	

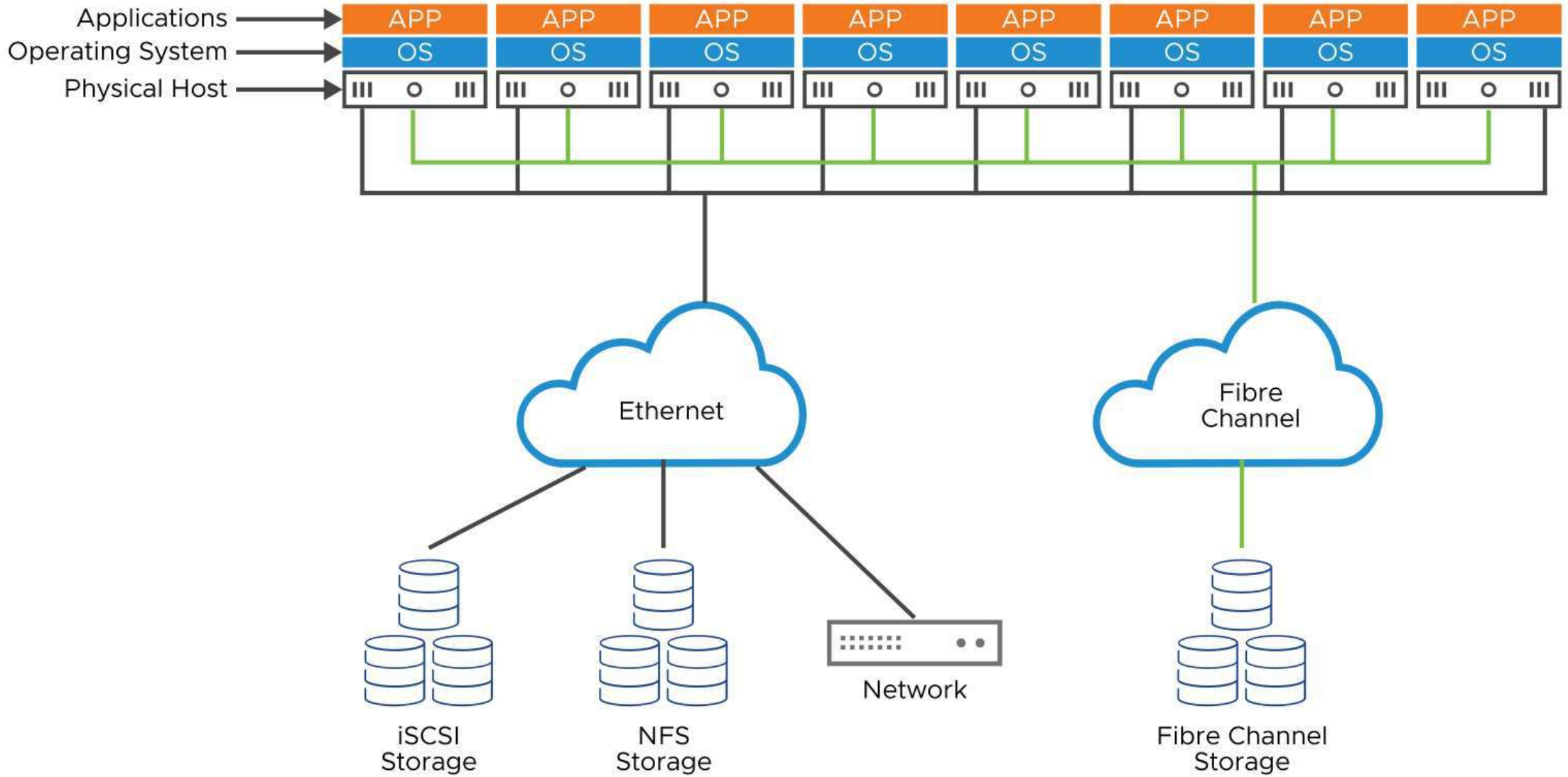
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Terminology (2)

Term	Definition
vSphere	Server virtualization product of VMware that combines the ESXi hypervisor and the vCenter Server management platform
Cluster	Group of ESXi hosts whose resources are shared by VMs
vSphere vMotion	Feature that supports the migration of powered-on VMs from host to host without service interruption
vSphere HA	Cluster feature that protects against host hardware failures by restarting VMs on hosts that are running normally
vSphere DRS	Cluster feature that uses vSphere vMotion to place VMs on hosts and ensure that each VM receives the resources that it needs

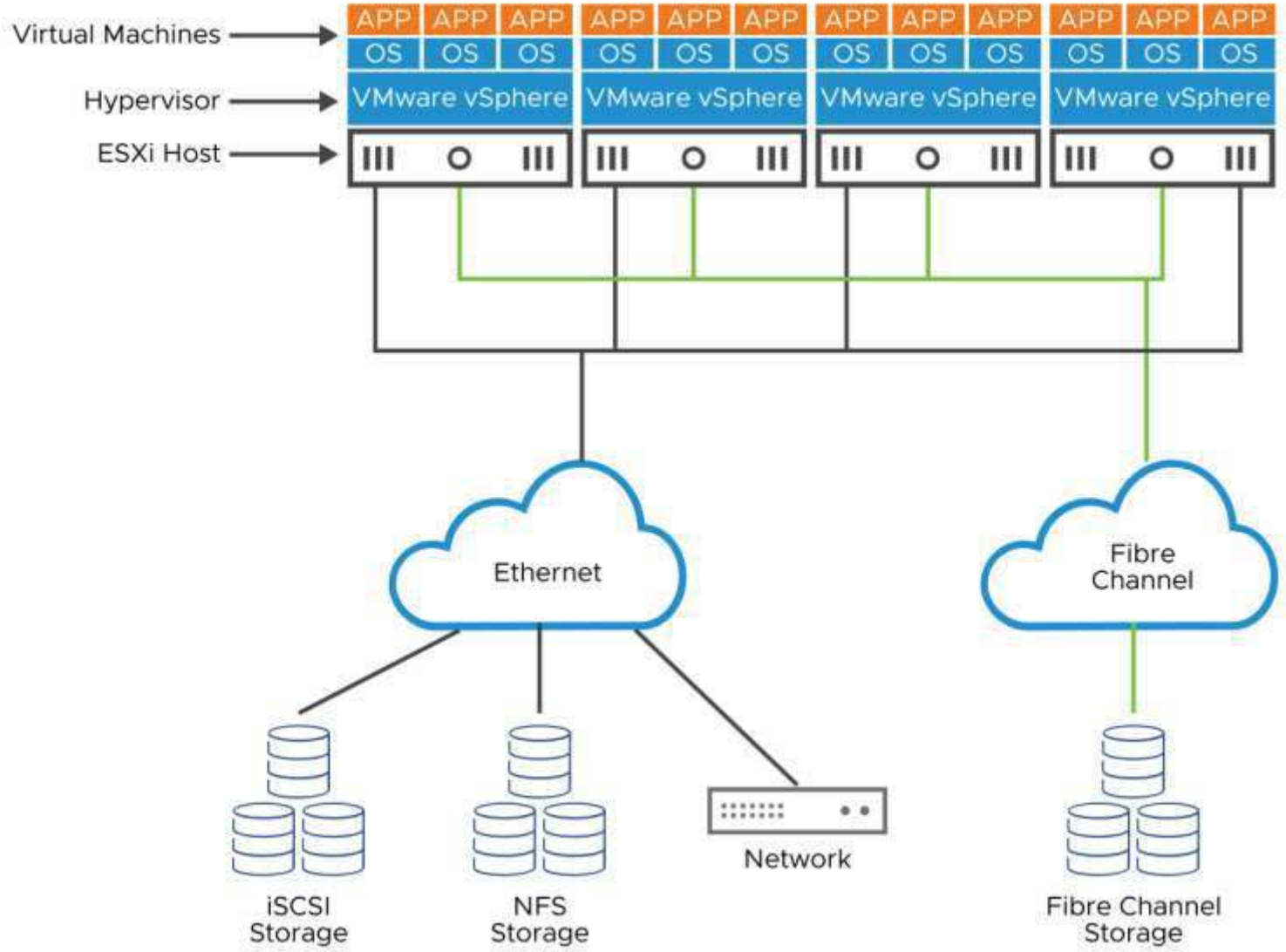
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Data Center Physical Infrastructure



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About the Virtual Infrastructure



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About Virtual Machines

A virtual machine (VM) is a software representation of a physical computer and its components.



The virtual machine includes the following components:

- Guest operating system
- VMware Tools
- Virtual resources, such as:
  - CPU and memory
  - Network adapters
  - Disks and controllers
  - GPUs

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Benefits of Using Virtual Machines

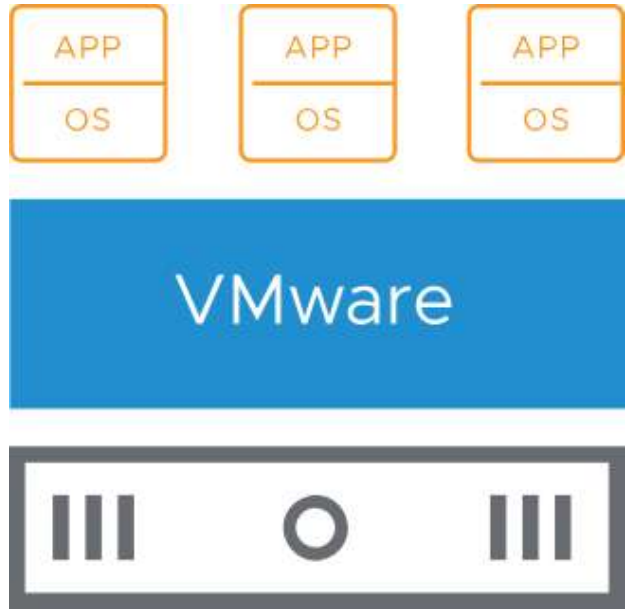
Physical machines have the following constraints:

- Difficult to move or copy
- Bound to a specific set of hardware components
- Often have a short life cycle
- Require personal contact to upgrade hardware



Virtual machines provide the following benefits:

- Easy to move or copy
- Independent of physical hardware because VMs are encapsulated into files
- Isolated from other VMs running on the same physical hardware
- Insulated from physical hardware changes

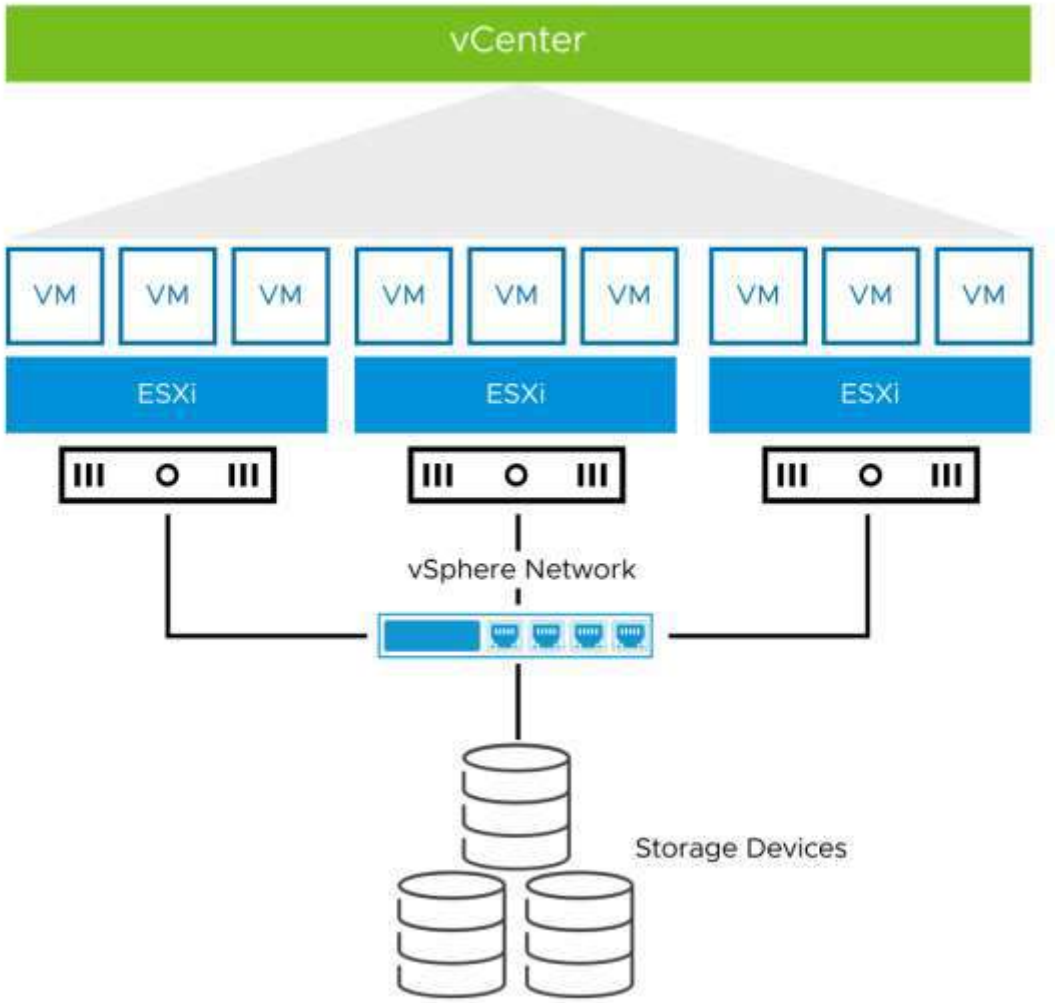


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About vSphere

vSphere is the virtualization platform that includes two core administrative components for running virtual machines:

- ESXi: Hypervisor on which you run virtual machines
- vCenter: Central administration platform for ESXi hosts, virtual machines, storage, and networking

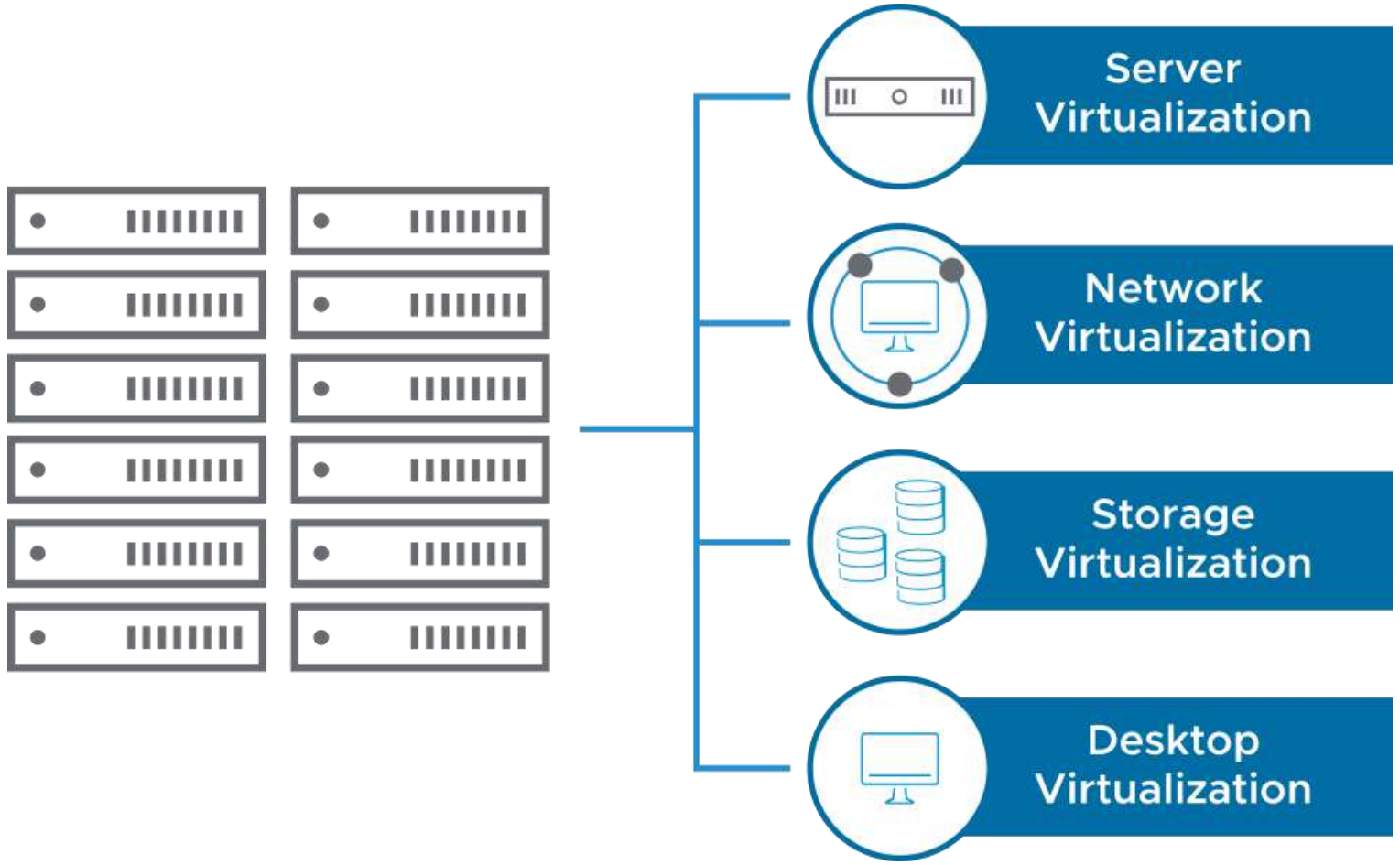


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Types of Virtualization

Virtualization is the process of creating a software-based representation of a physical unit, such as a server, desktop, network, or storage device.

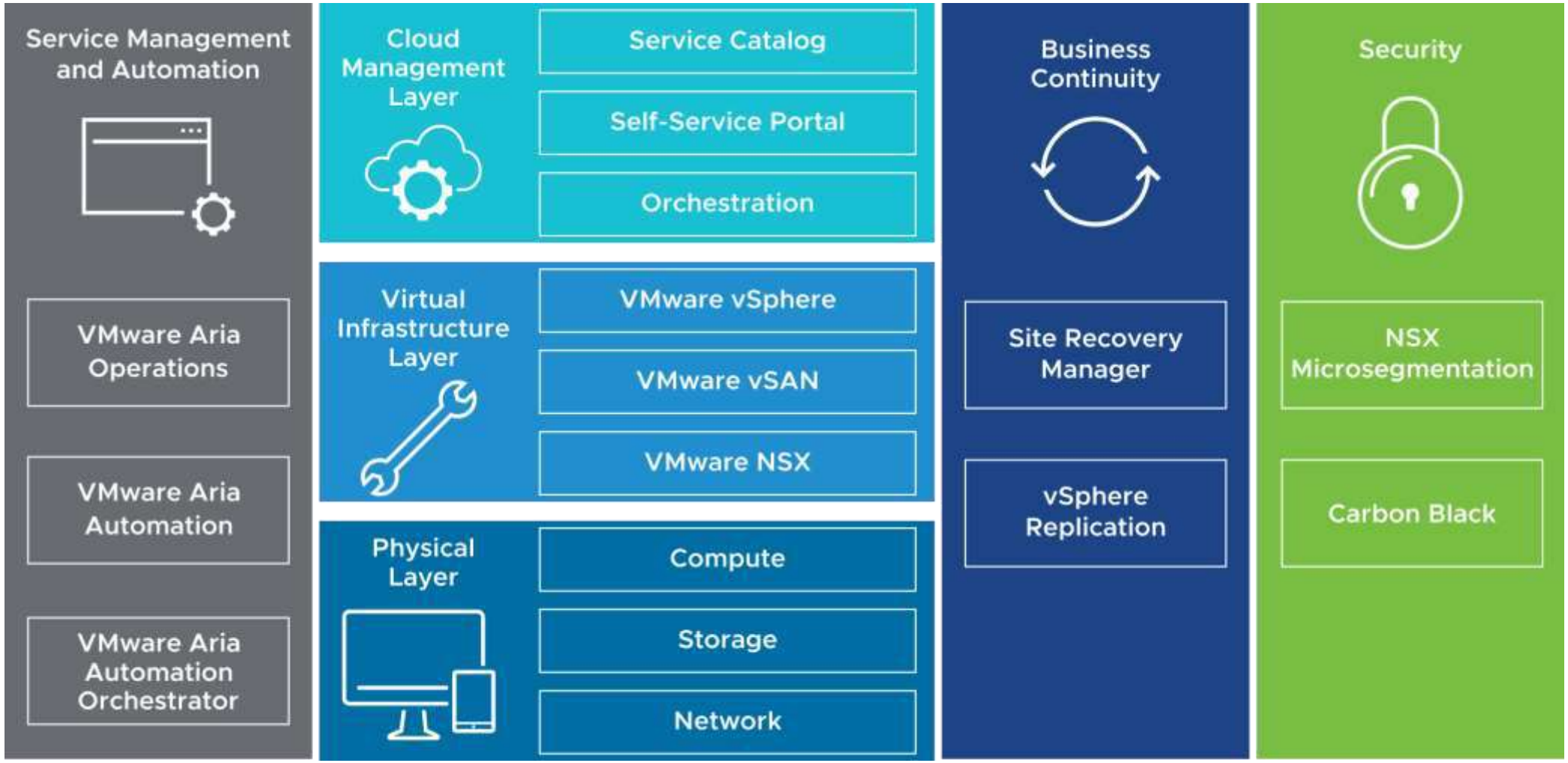
Virtualization is the single most effective way to reduce IT expenses while boosting efficiency and agility for all business sizes.



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About the Software-Defined Data Center

In a software-defined data center (SDDC), all infrastructure is virtualized, and the control of the data center is automated by software. vSphere is the foundation of the SDDC.

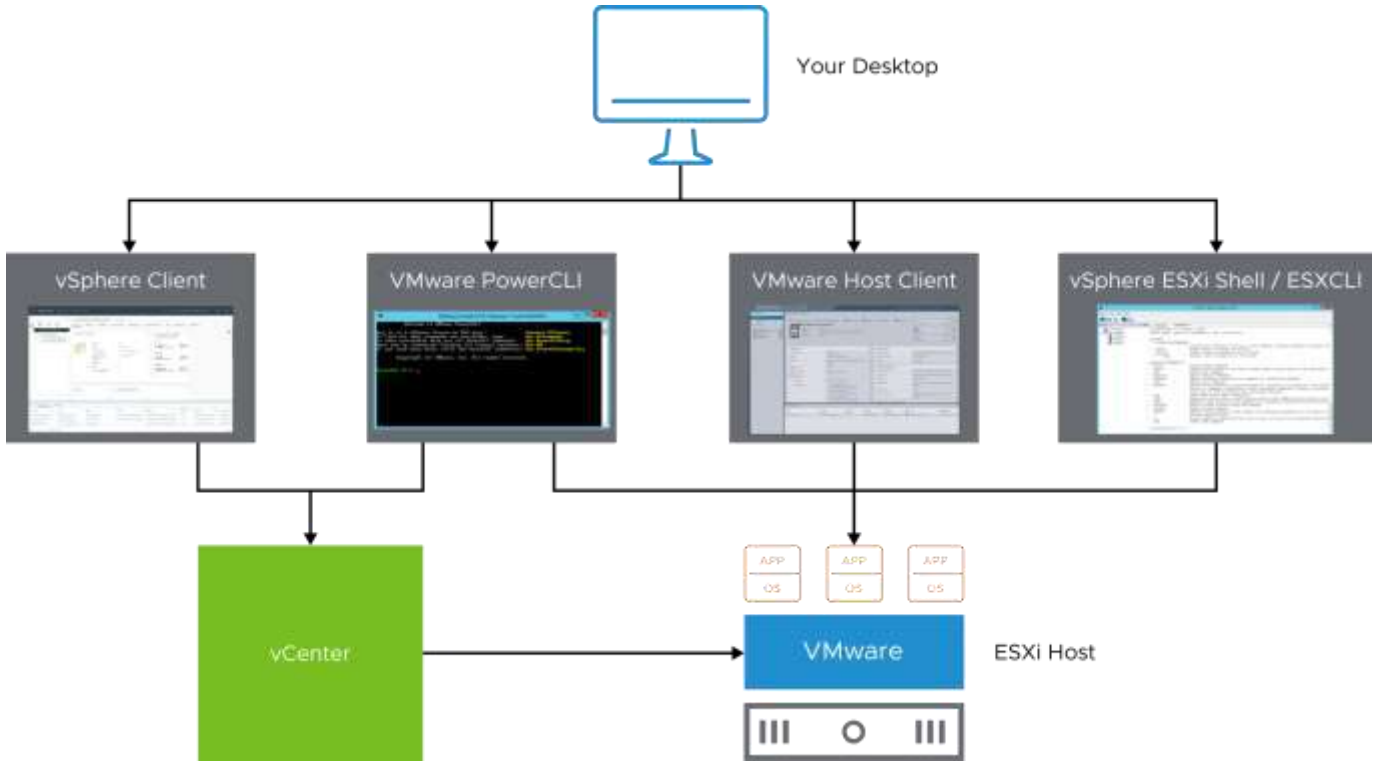


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vSphere User Interfaces

You can use the vSphere Client, PowerCLI, VMware Host Client, vSphere ESXi Shell and ESXCLI to interact with the vSphere environment.

For information about ports and protocols, see https://ports.vmware.com.



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